Surface Area of a Right Circular Cylinder

(i) Curved surface area

Curved surface area (C.S.A.) of a right circular cylinder

$$\begin{aligned} &= \text{Area of the corresponding rectangle} \\ &= l \times b \\ &= 2\pi r \times h \quad (\because l \text{ is the circumference} \\ &= 2\pi rh \quad \text{of the base, } b \text{ is the height) [see Fig. 7.5]} \end{aligned}$$

C.S.A. of a right circular cylinder $= 2\pi rh$ sq. units.

(ii) Total surface area

Total surface area refers to the sum of areas of the curved surface area and the two circular regions at the top and bottom.

That is, total surface area (T.S.A.) of right circular cylinder

$$\begin{aligned} &= \text{C.S.A + Area of top circular region} \\ &\quad \text{+Area of bottom circular region.} \\ &= 2\pi rh + \pi r^2 + \pi r^2 \quad \text{(Refer Fig.7.4)} \\ &= 2\pi rh + 2\pi r^2 \\ &= 2\pi r\left(h + r\right) \end{aligned}$$

T.S.A. of a right circular cylinder $= 2\pi r(h + r)$ sq. units

Note

- We always consider $\pi = \frac{22}{7}$, unless otherwise stated.
- The term 'surface area' refers to 'total surface area'.

Example 7.1 A cylindrical drum has a height of 20 cm and base radius of 14 cm. Find its curved surface area and the total surface area.

Solution Given that, height of the cylinder $h = 20$ cm ; radius $r = 14$ cm

Now, C.S.A. of the cylinder $= 2\pi rh$ sq. units

$$\text{C.S.A. of the cylinder} = 2 \times \frac{22}{7} \times 14 \times 20 = 2 \times 22 \times 2 \times 20 = 1760 \text{ cm}^2$$

$$\begin{aligned} \text{T.S.A. of the cylinder} &= 2\pi r\left(h + r\right) \text{ sq. units} \\ &= 2 \times \frac{22}{7} \times 14 \times \left(20 + 14\right) = 2 \times \frac{22}{7} \times 14 \times 34 \\ &= 2992 \text{ cm}^2 \end{aligned}$$

Therefore, C.S.A. = 1760 cm^2 and T.S.A. = 2992 cm^2

10th_Maths_Chapter 7_English.indd 271 12/3/2024 2:59:27 PM